Geochimica et Cosmochimica Acta xxx (xxxx) xxx

Contents lists available at ScienceDirect

Geochimica et Cosmochimica Acta

journal homepage: www.elsevier.com/locate/gca

Diagenetic fluid overprinting of micrite stable isotopes: implications for paleo-elevation reconstruction☆

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ARTICLE INFO

Co-Guest Editor: Hongqiang Yang

Keywords:
Tibetan Plateau
Lacustrine carbonate micrites
Paleo-elevation
Diagenesis
Calcium isotopes

ABSTRACT

Lacustrine carbonate micrite is commonly treated as a reliable archive of surface-water oxygen isotopes and is widely used for paleo-elevation reconstruction in orogenic belts. However, recent evidence for shallow-burial dolomitization in lacustrine micrites raises concern that bulk samples may not preserve a single primary signal. Here we evaluate the fidelity of geochemical signals preserved in the micritic carbonates of the Eocene Gerze Basin in central Tibet. The geochemical compositions of calcite and dolomite fractions in bulk samples were isolated using sequential acetic-acid leaching and show one primary or least altered calcite end-member (Calcite EM) and two diagenetic dolomite end-members (Dolomite EM1 and EM2). Numerical simulation suggests that the higher and more variable $\delta^{44/40}$Ca values in Dolomite EM2 formed under fluid-buffered conditions. In contrast, Dolomite EM1, with moderately positive $\delta^{44/40}$Ca values, is a product of sediment-buffered dolomitization. Accounting for this diagenetic alteration, each 10 % increase in the proportion of Dolomite EM2 in bulk carbonates would increase the reconstructed paleo-elevation by 0.5–0.8 km. These results show that Gerze Basin micrite is not a homogeneous recorder of lake-water δ^{18}O and that unrecognized dolomitization can substantially bias paleo-elevation reconstructions.

1. Introduction

The rise of mountain belts reflects deep Earth geodynamic processes and profoundly influences surface climate and biosphere systems (e.g., Molnar et al., 1993; Rowley and Currie, 2006; Wolf et al., 2022). Reconstructing paleo-elevation in orogenic belts commonly relies on stable isotopes preserved in continental carbonates such as lacustrine carbonates or pedogenic nodules (Poage and Chamberlain, 2001; Spicer et al., 2021; Ding et al., 2022; Quade et al., 2020). However, although lacustrine carbonates have been widely regarded as capable of preserving primary geochemical signatures, diagenetic overprinting can generate mineralogical, geochemical, and isotopic heterogeneity that may bias paleo-elevation estimates (Quade et al., 2020; Li et al., 2024a).

As the largest collisional plateau on Earth, the Tibetan Plateau hosted a longitudinal Eocene valley along the Bangong-Nujiang Suture Zone (Su et al., 2019; Su et al., 2020; Xiong et al., 2022; Fang et al., 2020; Ding et al., 2022; Li et al., 2024b), flanked by high Lhasa and Qiangtang terranes (Ma et al., 2026; Ding et al., 2022). The Gerze Basin, located in the western segment of this zone, has been the focus of ongoing debate regarding the paleoenvironmental conditions of the Eocene. Specifically, whether these conditions reflect low-elevation, near-sea-level environments or high-elevation topography remains unclear (Wei et al., 2016; Ingalls et al., 2020; Li et al., 2024a). A critical factor in this debate is the distinct diagenetic histories and isotopic signatures of

☆ This article is part of a special issue entitled: 'Geochemical Proxies_Zhang' published in Geochimica et Cosmochimica Acta.

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https://doi.org/10.1016/j.gca.2026.09.026
Received 30 December 2025; Accepted 21 September 2026
Available online 23 September 2026
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heterogeneous components, such as micritic matrix and sparite veins, in lacustrine carbonates. Although sparite veins with negative $\delta^{18}O$ values (~ −10 ‰) are regarded as products of early meteoric diagenesis (Ingalls et al., 2020), a recent study suggested that sparite veins formed from near-surface freshwater during late diagenesis (Li et al., 2024a). While the $\delta^{18}O$ value of the micritic component is ~ − 5 ‰, indicating the primary signal of open low-elevation lake water (Wei et al., 2016), Ingalls et al. (2020) argued that such positive values are due to the evaporative enrichment in a closed lake basin.

Despite its widespread use in paleo-elevation reconstructions, micrite exhibits substantial internal heterogeneity, most notably the frequent presence of mixed calcite-dolomite phases. These calcite-dolomite mixtures have been observed in numerous Cenozoic basins across Tibet (Cyr et al., 2005; Li et al., 2017; Li et al., 2019), but few studies have addressed the underlying processes responsible for this mineralogical mixing. Recent clumped-isotope analyses suggest that micrite in the Gerze Basin underwent shallow-burial dolomitization at temperatures of ~ 40–50 °C (Li et al., 2024a), which contrasts with surface conditions. Although dolomite formed during diagenesis is unlikely to preserve primary paleo-elevation signals, it remains uncertain whether coexisting calcite retains pristine isotopic compositions or has also undergone alteration.

In this study, we investigate Eocene lacustrine carbonates from the Gerze Basin using an integrated approach that combines petrological, geochemical, and modeling techniques. First, we analyzed samples using optical microscopy, cathodoluminescence (CL) imaging, scanning

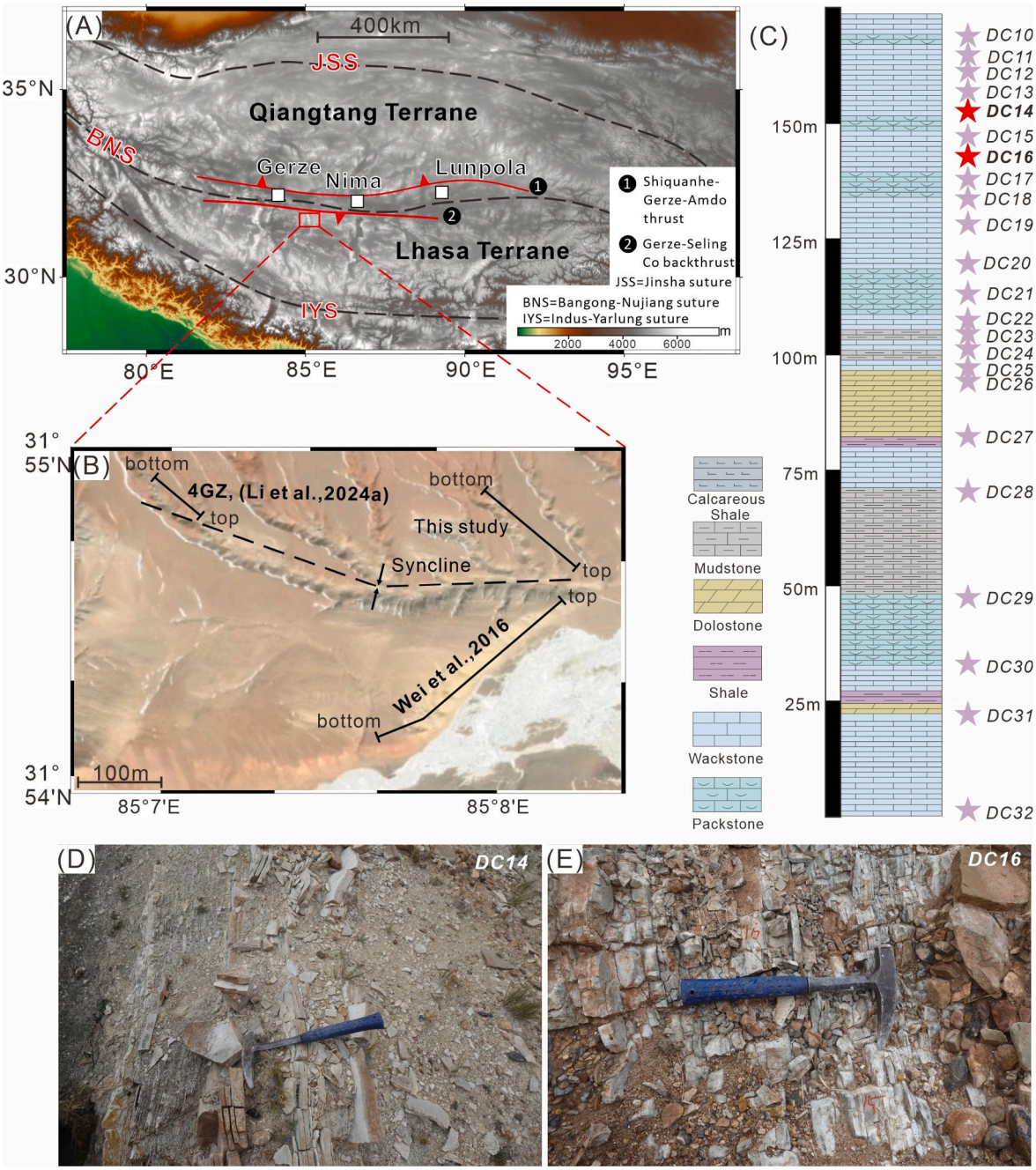


Fig. 1. Location map, stratigraphic column, and field photos of the research area. (A) Map showing the locations of study area, Gerze Basin on Tibetan Plateau; (B) Locations of previous and current research sites; (C) Stratigraphic column and sample locations in the Gerze Basin for this study; (D) Field photograph of sample DC14; (E) Field photograph of sample DC16.

electron microscopy (SEM), and X-ray diffraction (XRD) to assess mineralogical heterogeneity before proceeding with geochemical analyses. We then applied an optimized acetic-acid leaching method to separate calcite and dolomite fractions. Multi-proxy isotopic and elemental measurements, including $\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$, $\delta^{44/40}Ca$, $^{87}Sr/^{86}Sr$, Sr/Ca, and Mn/Sr, were performed to evaluate whether calcite preserves primary surface-water isotopic signatures and to assess the extent to which dolomitization alters $\delta^{18}O$ values. Calcium isotopes ($\delta^{44/40}Ca$) are particularly useful for tracing diagenesis (Gussone et al., 2005; Husson et al., 2015; Kimmig and Holmden, 2017; Ahm et al., 2018; Higgins et al., 2018; Lau and Hardisty, 2022; Fantle et al., 2020; Adiatma et al., 2024), and when combined with $\delta^{18}O_{carb}$, $^{87}Sr/^{86}Sr$, Sr/Ca, and Mn/Sr ratios, they can differentiate between fluid-buffered and sediment-buffered diagenesis while quantifying fluid isotopic compositions (Holmden, 2009; Lau et al., 2017; Fantle et al., 2020; Riechelmann et al., 2020; Adiatma et al., 2024; Liu et al., 2024). This integrated diagenetic evaluation allows us to assess the uncertainties associated with paleo-elevation reconstructions.

2. Geological background

The measured sedimentary section is located in the Gerze Basin within the western part of the Bangong-Nujiang Suture Zone. This suture zone is bounded from the Lhasa terrane to the south by the south-dipping Gerze-Seling Co backthrust and from the Qiangtang terrane to the north by the Shiquanhe-Gerze-Amdo thrust (Kapp et al., 2005; Hu et al., 2022; Fig. 1A). The studied section is situated on the northern limb of a syncline, less than 2 km from both the 4GZ section reported by Li et al. (2024a) on the northern limb and the section studied by Wei et al. (2016) on the southern limb (Fig. 1B).

The total thickness of the section is 175 m and mainly consists of lacustrine deposits which were defined as the Dingqinghu Formation (Fig. 1C). Light-gray marl wackestone beds (Fig. 1D), interbedded with gray intraclastic packstone (Fig. 1E), dominate the section, with subordinate dolostone and shale layers. Based on detrital zircon U-Pb ages (~42 Ma maximum depositional age), magnetostratigraphy, and index fossils such as the ostracod *Austrocypris posticaudata* and the charophyte *Gyrogona qianjiangica*, the depositional age is bracketed to 39.4–35.7 Ma in late Eocene (Wei et al., 2016).

3. Materials and methods

3.1. Sample preparation

A total of 24 rock samples were collected directly from the exposed outcrop in the Gerze Basin (Fig. 1D and E), with fresh surfaces obtained by shallow drilling of polished carbonate specimens. Thin sections were prepared and stained with Alizarin Red S to differentiate calcite from dolomite. Micrite samples were powdered using a SAESHIN® dental drill with a 500 μm bit at low rotational speeds with careful cleaning of the bit between samples to minimize contamination. In areas near veins or cement-filled pores, powders were carefully collected with a small knife to avoid mixing with vein material. Micro-drilling was performed using a 200 μm drill bit under an OLYMPUS® microscope.

3.2. Petrography and mineralogy

After microscopic analysis, thin sections were analyzed with a CL8200MK5 cathodoluminescence analyzer at Nanjing University to identify carbonate cement generations and replacement textures. Cathodoluminescence imaging was conducted at an accelerating voltage of 10.5 kV, a beam current of 250 μA, and a vacuum pressure of 0.03 mbar.

Mineralogical compositions of the micrite samples were determined by XRD using a Bruker D8 Advance diffractometer equipped with a LYNXEYE XE-T energy-dispersive detector at the State Key Laboratory of Critical Earth Material Cycling and Mineral Deposits at Nanjing University. Approximately 200 mg of powdered sample was analyzed over a 2θ range of 10°–80° with a step interval of 0.01° and a count time of 1 s per step. Semi-quantitative mineral abundances were calculated using the reference intensity ratio method.

To investigate the petrological characteristics of micrite at the micro–nano scale, scanning electron microscopy (SEM) with energy-dispersive spectroscopy (EDS) was performed using a Zeiss Sigma 500 instrument at the Key Laboratory of Surficial Geochemistry, Ministry of Education, Nanjing University. SEM imaging and EDS spot analyses were used to characterize carbonate mineral textures and provide semi-quantitative compositional constraints. The nominal electron-beam spot size was approximately 1–5 nm under the analytical conditions, although the effective EDS interaction volume varies with beam current, accelerating voltage, aperture, and sample composition. Samples were coated with a platinum film using a QUORUM Q150 sputter coater before analysis.

3.3. Carbonate isotope and elemental analyses

For bulk carbonate carbon and oxygen isotopes, about 1 mg of powdered sample was reacted with phosphoric acid at 70 °C for 24 min to determine $\delta^{13}C_{carb}$ and $\delta^{18}O_{carb}$ values using a Thermo Fisher Delta Plus XP Isotope Ratio Mass Spectrometer coupled with a Gas Bench continuous flow system at the State Key Laboratory of Critical Earth Material Cycling and Mineral Deposits, Nanjing University. The external reproducibility of $\delta^{13}C_{carb}$ and $\delta^{18}O_{carb}$ measurements, reported relative to VPDB, is better than ± 0.1 ‰ based on replicate measurements of the Chinese GBW00405 carbonate standard. Because calcite and dolomite may exhibit different phosphoric-acid fractionation behavior, potential mineral-specific offsets at 70 °C (Kim et al., 2015) were considered during cross-mineral comparison and temperature interpretation. The calibration procedure is summarized in Supplementary Section S1.

For bulk element, strontium and calcium isotopic analyses, approximately 10 mg of powdered carbonate was digested with 10 mL of 10 % ultrapure acetic acid for 24 h and then centrifuged to remove insoluble residues. A 100-μL aliquot of the supernatant was diluted for elemental concentration determinations (Ca, Sr, Mn, and other elements) by Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES), whereas the remaining supernatant from the same digestion was used for strontium and calcium isotope analyses. Strontium was purified by cation-exchange column chemistry using SR Resin (TrisKem, 50–100 μm), and $^{87}Sr/^{86}Sr$ was measured on a Thermo Fisher Scientific Triton Plus TIMS at the State Key Laboratory of Oil and Gas Reservoir Geology and Exploitation, Chengdu University of Technology. Calcium isotope ($\delta^{44/40}Ca$) analyses were performed on the same TIMS following Yang et al. (2026). A double-spike (^{42}Ca:^{43}Ca = 5:1; Liu et al., 2020) was added prior to column separation, and calcium was purified by cation-exchange column chemistry using AGMP-50 resin (Bio-Rad, 100–200 mesh) following Liu et al. (2022). The $\delta^{44/40}Ca$ values are reported relative to NIST 915a.

Analytical reproducibility was assessed using JDo-1, NBS987, NIST SRM 915a, and IAPSO seawater for monitoring elemental concentrations and Sr-Ca isotopes. Reproducibility was better than 0.2 wt% for Mg and Ca, 1.3 μg/g for Mn, 5.7 μg/g for Sr, 0.000028 for $^{87}Sr/^{86}Sr$, and 0.11 ‰ for $\delta^{44/40}Ca$. Full reference-material results are provided in Supplementary S2 and Fig. S1.

3.4. Sequential acetic acid leaching

Aliquots of the same powdered samples were analyzed using the bulk procedures described in Section 3.3 and the sequential acetic-acid leaching procedure described below. The bulk analyses provide integrated carbonate compositions, whereas sequential leaching resolves operationally defined calcite-rich and dolomite-rich fractions for end-member identification and subsequent mixing-model calculations.

To isolate the calcite and dolomite fractions from bulk samples, we

developed a two-step acetic acid leaching method, adapted from Swart and Melim (2000) and Lu et al. (2024a), Lu and Swart (2024b). Since lacustrine carbonate may contain clay minerals that could contaminate the carbonate fractions, we used low-concentration (< 15 % v/v) acetic acid to preferentially dissolve carbonate minerals while minimizing clay dissolution (Rongemaille et al., 2011).

Optimization experiments were conducted to determine the optimal acid concentration and leaching duration. Three representative samples were selected: DC12 (calcite-rich, 70.9 % calcite), DC14 (balanced calcite-dolomite, ~47.0 % each), and DC19 (dolomite-rich, 68.6 % dolomite). We tested 5 %, 10 %, and 15 % acetic acid over leaching intervals of 5 to 30 min. The experimental results show that the calcite fraction was not completely dissolved by 5 % acetic acid after 30 min (Fig. S2). This long-duration dissolution would further release calcium, strontium, and other geochemical signals from the clay fraction (Coimbra et al., 2021; Xu et al., 2024). In contrast, the leaching reaction of 15 % acetic acid is too rapid to retain sufficient calcite in the first residue (Fig. S3). Therefore, 10 % acetic acid, which provides a moderate dissolution rate in an optimum reaction time, was selected for the sequential leaching method to effectively separate dolomite and calcite fractions (Fig. S4).

Approximately 0.5 g of powdered sample was leached with acetic acid for 5 and 30 min to preferentially dissolve the calcite fraction. The first leachate (Leachate 1) was collected for elemental and Sr–Ca isotope measurements after 5 min. An aliquot of the parent residue (Residue 1) was used for XRD and $\delta^{13}C_{carb}$ and $\delta^{18}O_{carb}$ measurements. Then the remains of Residue 1 were dissolved for an additional 25 min to obtain the residual material (Residue 2), which is close to 100 % dolomite. Residues 1 and 2 were rinsed twice with deionized water and then oven-dried. Residue 2 was analyzed by XRD and measured for elemental concentrations, $\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$, $^{87}Sr/^{86}Sr$, and $\delta^{44/40}Ca$ values following the methods described in Section 3.3. The $\delta^{13}C_{carb}$ and $\delta^{18}O_{carb}$ values of calcite fractions were extrapolated by linear regression using the bulk sample and the two residual fractions (Residues 1 and 2). The sequential leaching workflow is shown in Fig. S5.

3.5. Models

3.5.1. Principal component and k-means analysis

Principal component analysis (PCA) was employed to reduce the dimensionality of the multi-proxy geochemical dataset (Abdi and Williams, 2010) and to objectively identify the dominant covariance patterns among $\delta^{18}O_{carb}$, $\delta^{13}C_{carb}$, $\delta^{44/40}Ca$, $^{87}Sr/^{86}Sr$, Sr/Ca, and Mn/Sr data. The first two principal components, which explain over 85 % of the total variance, were retained for visualization and further analysis. Additionally, k-means clustering was applied as an independent method for validating end-member proportions.

3.5.2. Bayesian Markov chain Monte Carlo (MCMC) model

To deconvolve the mixing process of micrite, a Bayesian Markov chain Monte Carlo (BayesMCMC; Norris, 1998; Rasmussen and Ghahramani, 2003) model was used to estimate the proportional contributions of different geochemical mineral end-members to the micrite. This model integrates $\delta^{13}C_{carb}$, $\delta^{18}O_{carb}$, $^{87}Sr/^{86}Sr$, and $\delta^{44/40}Ca$ isotopic data to probabilistically quantify mixing fractions, while accounting for end-member compositional variability and analytical uncertainties by:

$$\delta mix, i = \sum_{j=1}^{n} fj \times (\delta em, j, i + \varepsilon j, i) \quad (1)$$

The Bayesian inversion utilizes MCMC sampling with a Metropolis-Hastings algorithm to optimize model parameters (scaling factors α and isotopic corrections ε) by minimizing the weighted sum of squared residuals (Gelman et al., 2013; Tarantola, 2005). A total of 10,000 simulations were performed using Dirichlet distributions to constrain the model.

3.5.3. Water-rock reaction burial model

To quantitatively explore the influence of burial diagenesis, a generalized Monte Carlo water–rock reaction model that fundamentally expands the previous diagenetic framework of Veillard et al. (2019) and Lu et al. (2024c) was developed in this study. Unlike previous models that assumed fixed parameters, our model uses stochastic sampling of initial fluid compositions, water–rock ratios, and thermal histories to provide robust uncertainty bounds rather than fixed trajectories. This probabilistic approach includes mass-exchange dynamics, redefining the cumulative water–rock ratio (W/R) concept. Additionally, the model includes a temperature-triggered shift from calcite to dolomite fractionation to approximate oxygen-isotope evolution during burial recrystallization under dolomite-forming conditions. This model allows for a quantitative exploration of how fluid-source heterogeneity and burial gradients influence $\delta^{18}O_{carb}$ isotopic overprinting. The specific model parameters are provided in Table S1.

3.5.4. Diagenetic model

A diagenetic simulation model, following the framework of Ahm et al. (2018), was applied to estimate fluid interactions and early diagenetic processes responsible for micritic mixing effects. This framework assumes a single end-member fluid composition and is designed for early diagenetic settings rather than burial diagenesis (Akhtar et al., 2024; Nadeau et al., 2025). In the model, $\delta^{18}O$, $^{87}Sr/^{86}Sr$, and $\delta^{44/40}Ca$ and Sr/Ca are used to quantify diagenetic alteration under fluid-buffered to sediment-buffered conditions with different primary mineral and fluid combinations. The model removes original aragonite components due to mineral composition differences in lacustrine carbonates. Model parameters are provided in Table S2.

4. Results

4.1. Petrographic and mineralogical results

Petrographic analysis reveals that the micrite consists of granular intraclasts and a uniform matrix (Fig. 2A–B). Intraclasts are primarily composed of red-stained calcite, with minor dolomite, while the matrix is dominated by light yellow–brown dolomite, with limited red-stained calcite (Fig. 2A and B). In unstained thin sections, calcite appears mostly as sparite (Fig. 2C and D). Cathodoluminescence (CL) imaging shows weak to nearly non-luminescent behavior for both calcite and dolomite, indicating no significant CL contrast between the two phases (Fig. 2C–F).

X-ray diffraction (XRD) analysis shows that the lacustrine micrite consists mainly of dolomite and calcite, with quartz and clay minerals present in some samples (DC16, DC17, and DC23) at concentrations of 29.1 % to 52.5 % (Table 1). Except for samples DC08, DC12, DC20, DC23 and DC28, where dolomite contents range from 0.8 % to 37.1 % (mean = 23.9 %, carbonate-normalized), most micrite samples are dolomite-dominated (50.4 %–85 %, mean = 67.1 %, n = 12, carbonate-normalized). In contrast, all veins and vugs are calcite-dominated (Table 1).

Scanning electron microscopy (SEM) combined with energy-dispersive spectroscopy (EDS) reveals that calcite and dolomite occur as nano- to micrometer-scale particles, with mineral particles identified as pure calcite, low-magnesium transition minerals, and pure dolomite (Fig. 2G–H, EDS details are provided in Figs. S6 and S7).

4.2. Carbonate isotopic results

The $\delta^{13}C_{carb}$ values of bulk and micro-drilled samples range from −0.2 ‰ to − 3.9 ‰, and are statistically similar to those of calcite and dolomite fractions, with no significant systematic differences (Fig. 3A). Median $\delta^{13}C_{carb}$ values are consistent with previous studies (Wei et al., 2016; Li et al., 2024a), although some samples show more negative values.

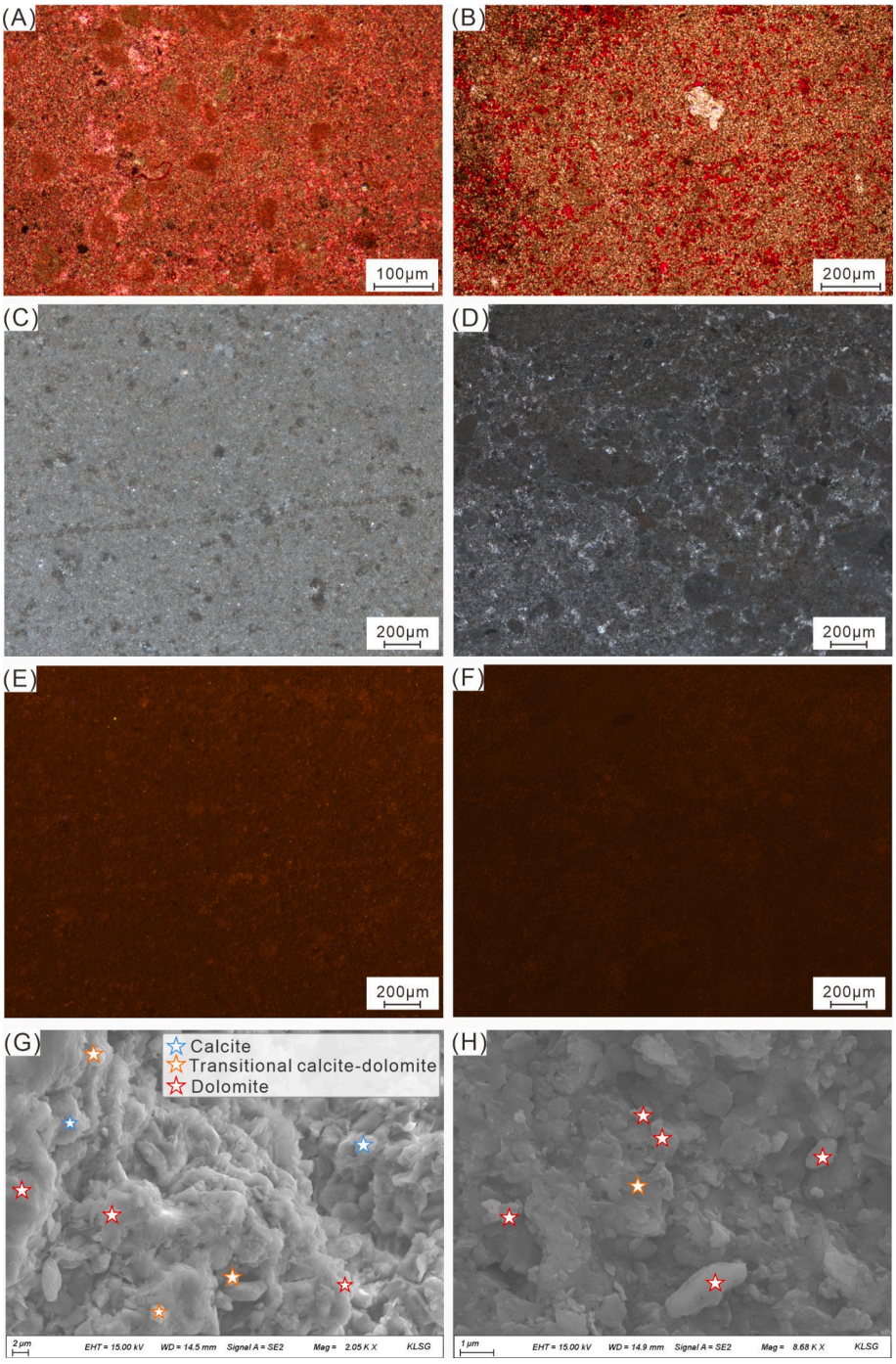


Fig. 2. Petrological features of lacustrine carbonate samples. Alizarin Red S-stained thin section images of samples (A) DC08 and (B) DC20; Microscopic images of samples (C) DC13 and (D) DC14; Cathodoluminescence (CL) images of samples (E) DC13 and (F) DC14; SEM images of samples (G) DC14 and (H) DC26 (EDS details are provided in Fig. S6-S7).

In contrast, $\delta^{18}O_{carb}$ values display distinct patterns after leaching. Bulk and micro-drilled samples range from –2.4 to –7.8 ‰, similar to those of calcite (–0.9 ‰ to –6.8 ‰), but more positive than dolomite values (–4.2 ‰ to –13.7 ‰; Fig. 3B). Values from Wei et al. (2016) are similar to our bulk, micro-drilled and calcite samples, while Li et al. (2024a) overlap with our dolomite sample results (Fig. 3B).

Bulk and micro-drilled $^{87}Sr/^{86}Sr$ ratios range from about 0.7090 to 0.7097, with leached calcite yielding lower ratios (0.7085–0.7090) compared to dolomite (0.7088–0.7096; Fig. 3C). $\delta^{44/40}Ca$ values range from 0.09 ‰ to 1.17 ‰ for bulk and micro-drilled samples, with dolomite generally exhibiting higher values (–0.03 ‰ to 1.14 ‰, median = 0.56 ‰) compared to calcite (–0.11 ‰ to 0.59 ‰, median = 0.17 ‰; Fig. 3D).

4.3. End-member division and quantization

A $\delta^{44/40}Ca$-$\delta^{18}O_{carb}$ cross-plot identifies three geochemical end-members, including a calcite end-member (Calcite EM) and two dolomite end-members (Dolomite EM1 and Dolomite EM2) (Fig. 4A, Table 2). Calcite EM has relatively positive $\delta^{18}O_{carb}$ (–6.84 ‰ to –0.92 ‰) and moderate $\delta^{44/40}Ca$ (–0.11 ‰ to 0.59 ‰), while Dolomite EM1 exhibits $\delta^{44/40}Ca$ values of 0.45 ‰ to 0.82 ‰ and $\delta^{18}O_{carb}$ values

Table 1
XRD results of lacustrine micrites in the Gerze Basin.

Sample	Fabric	Calcite	Dolomite	Quartz	Clay
		wt %, weight percentage			
DC08	micrite	62.7	37.1	0.1	0.1
DC09	micrite	43.8	54.4	0.2	1.6
DC10	micrite	23.5	71.3	3.2	2.0
DC11	micrite	13.5	76.5	4.9	5.1
DC11	vein	97.3	2.7	0.0	0.0
DC12	micrite	70.9	16.9	3.4	8.8
DC12	vein	95.2	4.8	0.0	0.0
DC14	micrite	46.5	47.3	0.3	5.9
DC14	calcite-fill vug	92.3	7.6	0.0	0.1
DC16	micrite	20.4	37.4	5.1	37.1
DC17	micrite	25.2	45.7	1.4	27.7
DC19	micrite	15.2	68.6	2.2	14.0
DC20	micrite	65.7	34.3	0.0	0.0
DC21	micrite	30.5	68.0	1.5	0.0
DC22	micrite	28.5	64.5	2.1	4.9
DC23	micrite	47.1	0.4	27.9	34.6
DC23	calcite-fill vug	98.5	1.5	0.0	0.0
DC24	micrite	24.7	73.7	1.3	0.3
DC24	vein	99.5	0.5	0.0	0.0
DC26	micrite	42.5	44.0	0.0	13.5
DC27	micrite	35.7	63.1	0.0	1.2
DC28	micrite	66.4	26.4	1.0	6.2

between –5.03 ‰ and –4.25 ‰. Dolomite EM2 is characterized by higher and more variable $\delta^{44/40}$Ca (–0.03 ‰ to 1.14 ‰) and lower $\delta^{18}O_{carb}$ (–13.70 ‰ to –8.45 ‰; Table 2).

Principal component analysis (PCA) and k-means clustering assign over 85 % of samples to these end-member groups, confirming the robustness of the three-component system (Fig. 4B). The corresponding loadings (Fig. 4C) show that PC1 (64.9 %) is dominated by $\delta^{44/40}$Ca, $^{87}Sr/^{86}Sr$, Sr/Ca and Mn/Sr, whereas $\delta^{18}O_{carb}$ and $\delta^{13}C_{carb}$ contribute to PC2, collectively driving the separation of the three end-members. Bayesian Monte Carlo Markov chain (BayesMCMC) modeling indicates that these three end-members explain the bulk micrite compositions well (Fig. 5A–D), with two- or four-end-member models failing to converge (Figs. S8 and S9). The results show that Dolomite EM2 predominantly contributes to the micrite matrix, comprising approximately 90 % ± 5 % of the mixture (Fig. 5E), while Calcite EM and Dolomite EM1 account for smaller fractions (10 % ± 5 % and 3 % ± 2 %, respectively). Sensitivity analysis confirms the dominance of Dolomite EM2, which accounts for at least 75 % of the micrite matrix (Fig. S10).

5. Discussion

We divide the paragenetic sequence into Stage 1, involving primary lacustrine deposition and early neomorphic modification (Calcite EM) and limited dolomitization (Dolomite EM1), and Stage 2, characterized

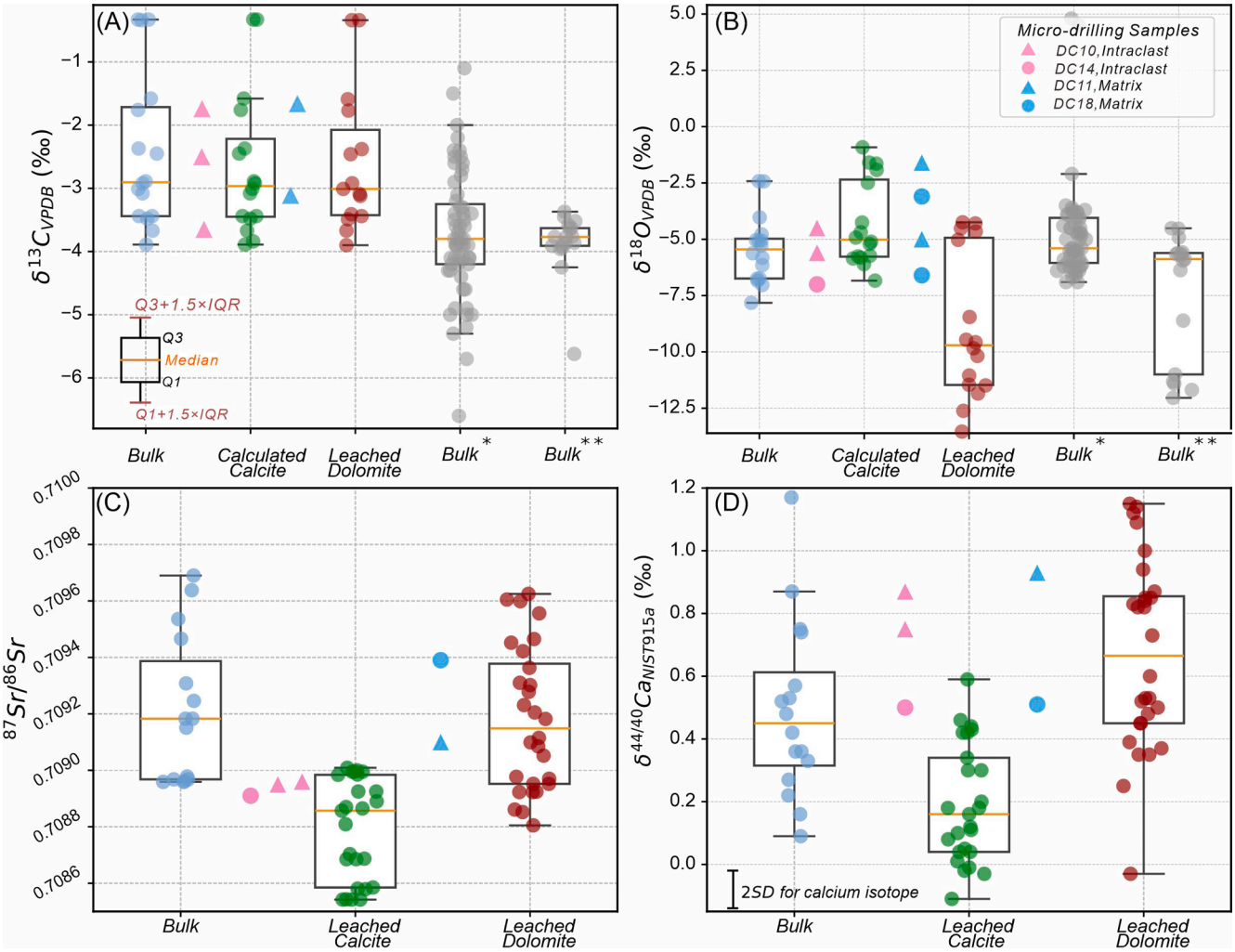


Fig. 3. Box plots of multiple geochemical results for different fabrics. (A) Box plot of $\delta^{13}C_{carb}$; (B) Box plot of $\delta^{18}O_{carb}$; (C) Box plot of $^{87}Sr/^{86}Sr$; (D) Box plot of $\delta^{44/40}$Ca. Bulk* and Bulk** in (A) and (B) refer to data from Wei et al. (2016) and Li et al. (2024a), respectively. Note that the calculated calcite values were derived using linear regression methods, and the leached results were analyzed directly.

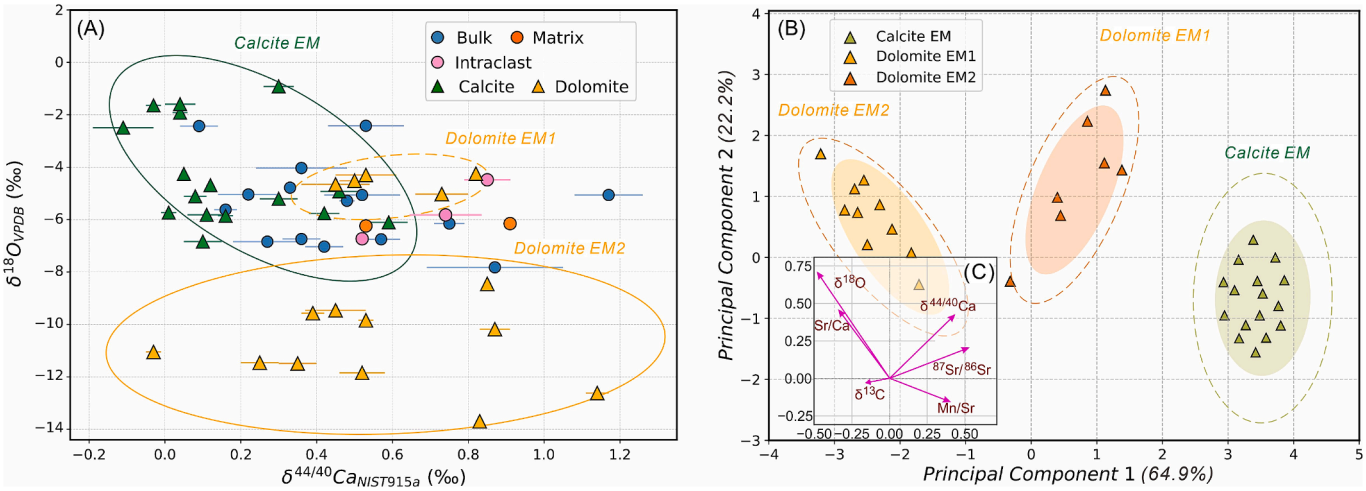


Fig. 4. (A) Cross-plot of $\delta^{44/40}$Ca vs. $\delta^{18}O_{carb}$; (B) and (C) Principal component analysis (PCA) results of $\delta^{44/40}$Ca, $\delta^{18}O_{carb}$, $\delta^{13}C_{carb}$, $^{87}Sr/^{86}Sr$, Sr/Ca, and Mn/Sr, showing three end-member clusters. The clusters are estimated using the k-means algorithm.

Table 2
Isotopic ranges of $\delta^{44/40}$Ca, $^{87}Sr/^{86}Sr$, $\delta^{13}C$ and $\delta^{18}O$ for calcite and dolomite end-members.

End-member	$\delta^{44/40}$Ca	$^{87}Sr/^{86}Sr$	$\delta^{13}C$	$\delta^{18}O$
	SRM915a/‰		VPDB/‰	VPDB/‰
Calcite EM	−0.11 to 0.59	0.7085 to 0.7090	−3.9 to −0.3	−6.84 to −0.92
Dolomite EM1	0.45 to 0.82	0.7088 to 0.7089	−3.4 to −0.3	−5.03 to −4.25
Dolomite EM2	−0.03 to 1.14	0.7089 to 0.7096	−3.9 to −0.3	−13.70 to −8.45

by pervasive dolomitization (Dolomite EM2). Calcite EM is interpreted as the earliest carbonate phase because (1) Calcite EM occurs as red-stained intraclasts or irregular patchy micrites without luminescence (Fig. 2A–D and E–F); (2) minerals in Calcite EM are subhedral granular or irregular (Fig. 2G–H); (3) the $\delta^{13}C_{carb}$ and $\delta^{18}O_{carb}$ values in Calcite EM are similar to those of modern Tibet lake calcite (Fig. 6A). Within Calcite EM, the high-Sr/Ca subset is interpreted as the least-altered calcite precursor, whereas the low-Sr/Ca subset records subsequent recrystallization. Dolomite EM1 formed through limited dolomitization because (1) it partially replaces Calcite EM (Fig. 2A); (2) it consists of finely crystalline dolomite which has dull luminescence (Fig. 2A, E and G–H). In contrast, the non-luminescent Dolomite EM2 (Fig. 2F) yields planar-euhedral to subhedral microcrystalline dolomite (4–10 μm), which indicates shallow-burial dolomitization. These petrographic characteristics, with no visible crystals under an optical microscope, favor a shallow-burial paragenetic sequence rather than deep-burial or hydrothermal alteration (Jiang et al., 2025).

5.1. Early diagenetic pathways of lacustrine micrite

This shallow-burial paragenetic framework provides the basis for evaluating the diagenetic pathways responsible for the contrasting geochemical signatures of Calcite EM, Dolomite EM1, and Dolomite EM2. A central issue is whether the observed $\delta^{18}O_{carb}$ differences among these end-members reflect temperature-dependent fractionation, fluid isotopic composition, kinetic effects, or different degrees of water–rock interaction.

5.1.1. Controls on the negative oxygen-isotope shift in Dolomite EM2

Previous studies suggest that shallow-burial diagenesis typically occurs at temperatures below ~ 70 °C (Bathurst, 1972; Worden and Burley, 2003; Pichat et al., 2016; Kim et al., 2019), consistent with the clumped-isotope temperatures of 40–50 °C reported from the adjacent 4GZ section (Li et al., 2024a). To test whether temperature-dependent fractionation alone can account for the negative $\delta^{18}O_{carb}$ shift in Dolomite EM2, a simplified burial water–rock interaction model was applied by assuming the $\delta^{18}O$ values (−10 ‰ to 0 ‰) represent lake water (Henderson et al., 2010; Wu et al., 2022). Within this shallow-burial temperature range, the model predicts that the mean $\delta^{18}O_{carb}$ value should be − 3.4 ± 2.2 ‰ (Fig. 6B), which is substantially more positive than the measured values of Dolomite EM2. Moreover, to account for the measured value of Dolomite EM2, the temperature should increase to 225 °C (Fig. 6C), which is much higher than the burial history in the studied area. Therefore, the increasing temperature cannot explain the negative oxygen-isotope shift recorded by Dolomite EM2.

A kinetic origin for the negative $\delta^{18}O_{carb}$ shift can also be tested using the coupled $\delta^{44/40}$Ca-Sr/Ca systematics. Variations in carbonate precipitation rate can drive kinetic fractionation in both oxygen and calcium isotopes (DePaolo, 2011; Wang et al., 2021, 2023). Under kinetic control, $\delta^{44/40}$Ca and Sr/Ca values are expected to show an inverse relationship with a slope of approximately − 1.1, as predicted for kinetic calcite precipitation (Tang et al., 2008). The Dolomite EM2 trend instead has a much steeper slope of − 4.1 (Fig. 7A), which is inconsistent with the expected kinetic-fractionation relationship. Kinetic effects are therefore unlikely to be the primary control on the observed negative oxygen-isotope shift.

Because neither temperature-dependent fractionation nor kinetic effects can account for the observed trend, the negative $\delta^{18}O_{carb}$ shift in Dolomite EM2 is best explained by interaction with an ^{18}O-depleted diagenetic fluid. This fluid must have had $\delta^{18}O$ values lower than those of normal lake water (−10 ‰ to 0 ‰), plausibly reflecting meteoric recharge, snowmelt, or glacial runoff in a high-altitude lacustrine setting (Shen and Poulsen, 2019; Li et al., 2019; Liu et al., 2026).

5.1.2. Fluid-buffered trend in Dolomite EM2

Dolomite EM2 accounts for most of the micrite matrix (~90 %; Fig. 5E) and is most closely associated with Stage 2 dolomitization. The geochemical composition of Dolomite EM2 therefore represents the fluid-buffered pathway. In a fluid-buffered system, increasing water–rock interaction is expected to increase $\delta^{44/40}$Ca values and decrease Sr/Ca ratios, while $^{87}Sr/^{86}Sr$ shifts toward the composition of the interacting pore fluid (Banner and Hanson, 1990; Banner, 1995; Jacobson and Holmden, 2008; Ahm et al., 2018; Higgins et al., 2018; Kimmig et al., 2021; Liu et al., 2024). From Calcite EM to Dolomite EM2, $\delta^{44/40}$Ca increases from 0.10 ± 0.30 ‰ to 0.70 ± 0.50 ‰, Sr/Ca

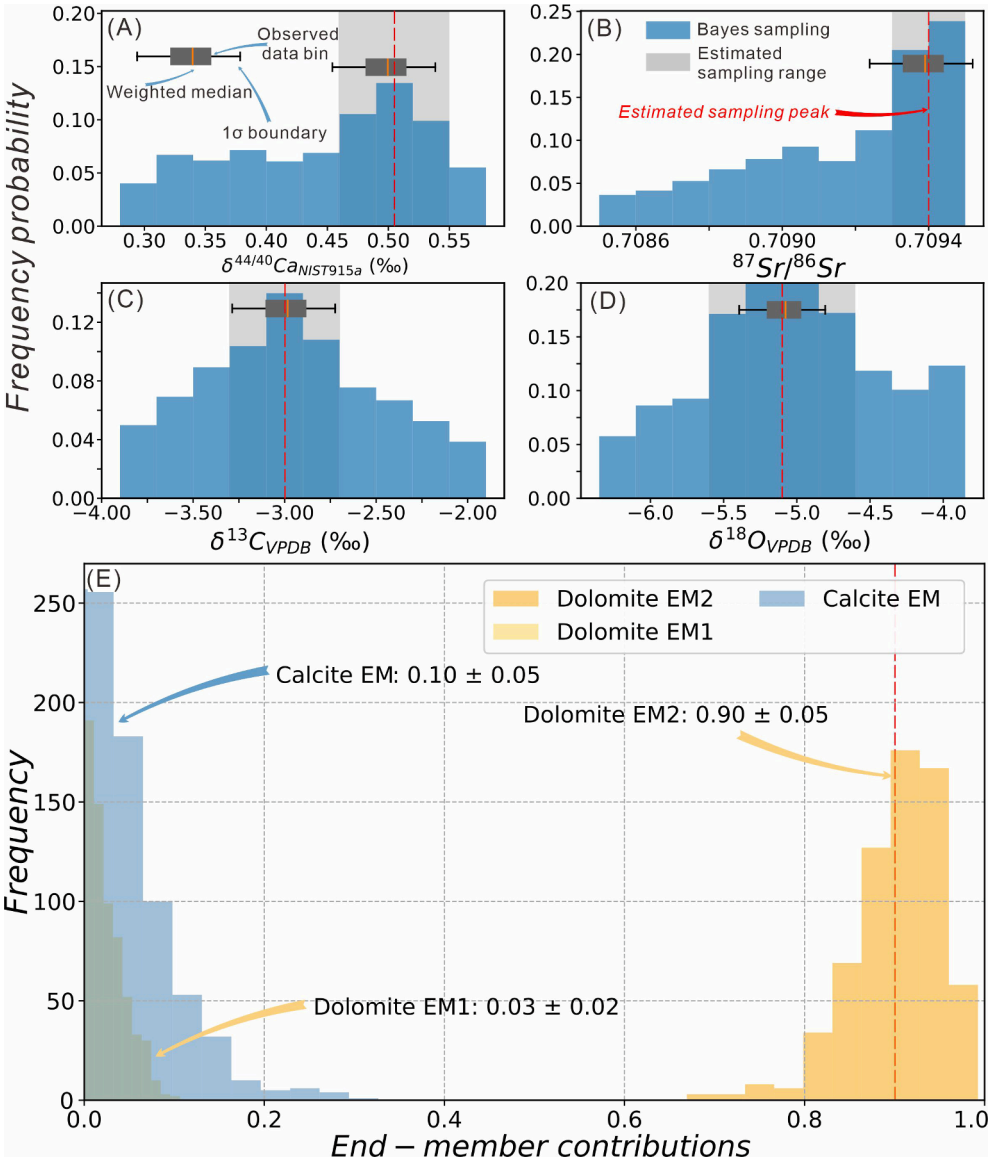


Fig. 5. BayesMCMC modeling of three end-member mixing. (A) Comparison of Bayesian inverse sampling with observed $\delta^{44/40}$Ca; (B) Comparison of Bayesian inverse sampling with observed $^{87}Sr/^{86}Sr$; (C) Comparison of Bayesian inverse sampling with observed $\delta^{13}C_{carb}$; (D) Comparison of Bayesian inverse sampling with observed $\delta^{18}O_{carb}$; (E) MCMC forward calculation of end-member contributions. Note that marginal posterior summaries do not necessarily sum exactly to unity because they were calculated independently for each end-member.

decreases from 2.3 ± 0.3 to 0.7 ± 0.3 (Fig. 7A), and $^{87}Sr/^{86}Sr$ ratios shift from 0.7086 toward 0.7096 (Fig. 7B). These paired isotopic and elemental changes reflect increasing fluid influence during the more pervasive dolomitization.

The systematic covariation between $^{87}Sr/^{86}Sr$ and $\delta^{44/40}$Ca further supports mixing between a lake-water-dominated precursor and an evolved diagenetic fluid (Fig. 7B). Carbonate precipitation can drive $\delta^{44/40}$Ca values upward through preferential removal of ^{40}Ca, whereas interaction with continental fluids can increase the $^{87}Sr/^{86}Sr$ ratio of the diagenetic fluid (Fantle and DePaolo, 2007; Shao et al., 2018; Rhodes et al., 2002; Akhtar et al., 2024). Accordingly, Calcite EM ($\delta^{44/40}$Ca ≈ 0.1 ‰, $^{87}Sr/^{86}Sr$ ≈ 0.7086, Sr/Ca ≈ 2.3) represents the lake-water-dominated end-member, whereas Dolomite EM2 ($\delta^{44/40}$Ca ≈ 0.76 ‰, $^{87}Sr/^{86}Sr$ ≈ 0.7096, Sr/Ca ≈ 0.7) approaches the diagenetic-fluid end-member. Therefore, the shallow-burial fluid $^{87}Sr/^{86}Sr$ value is estimated to be approximately 0.7096.

A subset of Calcite EM samples with low Sr/Ca ratios plots slightly away from the main $\delta^{44/40}$Ca-Sr/Ca trend (Fig. 7A). However, these Calcite EM samples with low Sr/Ca ratios remain aligned with the $\delta^{44/40}$Ca-$^{87}Sr/^{86}Sr$ and $\delta^{44/40}$Ca-Mn/Sr relationships (Fig. 7B–C), indicating continued recrystallization within the same fluid-buffered framework rather than a separate diagenetic pathway.

The $\delta^{44/40}$Ca, $^{87}Sr/^{86}Sr$, Sr/Ca, and Mn/Sr systematics define a semi-quantitative two-component framework in Calcite EM and Dolomite EM2. Calcite EM preserves a first-order approximation of the lake-water-dominated precursor, whereas Dolomite EM2 records greater interaction with an evolved shallow-burial diagenetic fluid. These relationships provide the boundary conditions for the quantitative modeling in Section 5.2.

5.1.3. *Sediment-buffered system in Dolomite EM1*

In contrast to the fluid-buffered trend between Calcite EM and Dolomite EM2, Dolomite EM1 records a sediment-buffered diagenetic pathway. Although Dolomite EM1 could be interpreted as primary lacustrine dolomite (e.g., Chang et al., 2020), this interpretation is difficult to reconcile with the petrographic and geochemical evidence.

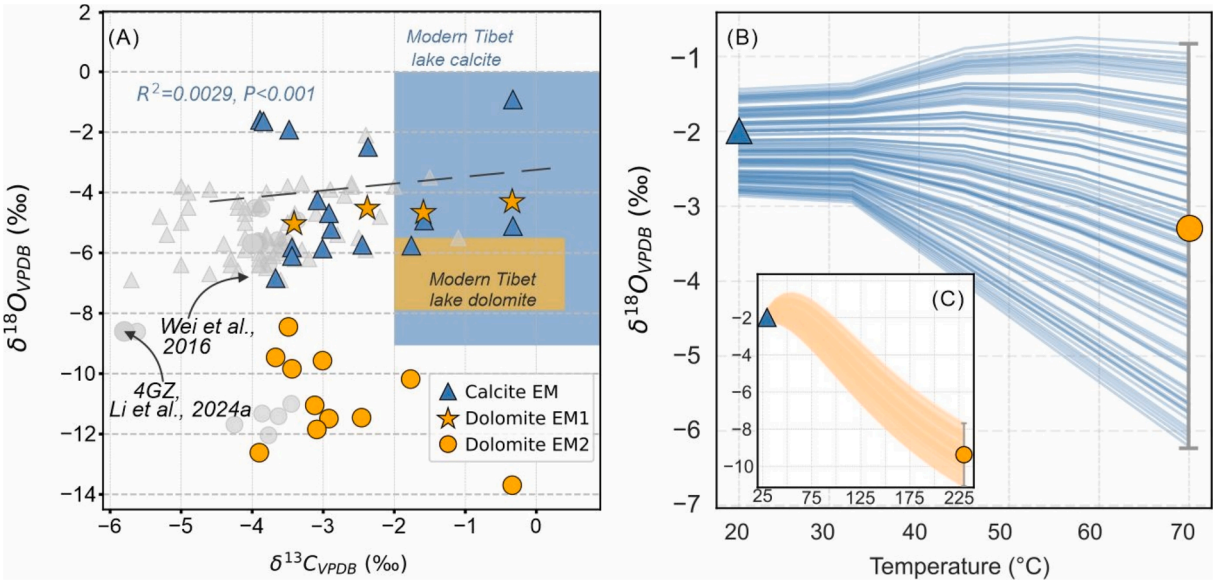


Fig. 6. (A) Cross-plot of $\delta^{13}C_{carb}$ vs. $\delta^{18}O_{carb}$ and water–rock reaction burial models for (B) 20 – 70° C and (C) 25 – 225 °C. The modern carbonate $\delta^{13}C_{carb}$ and $\delta^{18}O_{carb}$ values were compiled from Wu et al. (2022).

Modern primary lacustrine dolomite commonly requires extreme evaporation, microbial mediation, or hydrothermal input (Sánchez-Román et al., 2011), none of which is evident in the Gerze Basin micrites. Dolomite EM1 has relatively high Sr concentrations and Sr/Ca ratios, and its $^{87}Sr/^{86}Sr$ values (0.7088–0.7089) remain close to the less radiogenic precursor carbonate. This combination is interpreted as retention of a Sr-rich precursor signal during limited early dolomitization rather than as evidence for an externally buffered fluid. Together with clumped-isotope temperatures of 40–50 °C for dolomitic micrite in the adjacent 4GZ section (Li et al., 2024a), these features indicate that Dolomite EM1 is a diagenetic replacement product formed under sediment-buffered conditions.

The sediment-buffered interpretation is supported by the coupled Ca-Sr isotope and Sr-Mn elemental characteristics of Dolomite EM1. Dolomite EM1 has positive $\delta^{44/40}Ca$ values and relatively high Sr/Ca ratios, a pattern that follows sediment-buffered behavior and suggests preservation of the calcite-precursor Ca-isotope signature (Ahm et al., 2018; Higgins et al., 2018; Liu et al., 2024). The $\delta^{44/40}Ca$ and $^{87}Sr/^{86}Sr$ values of Dolomite EM1 depart from the fluid-buffered trend defined by Calcite EM and Dolomite EM2, indicating limited exchange with an evolved external fluid (Fig. 7B). The Mn/Sr and Sr/Ca values of Dolomite EM1 are also closer to those of high-Sr/Ca Calcite EM than to Dolomite EM2 (Fig. 7D).

A sediment-buffered Sr-Ca isotope signature does not preclude an external Mg supply. In lacustrine settings, the Mg supply could have been provided by evaporatively concentrated, relatively Mg-rich brines delivered by density-driven reflux (Warren, 2000; Al-Helal et al., 2012; Manche and Kaczmarek, 2023). Such reflux could introduce Mg into the sediment-buffered domain without fully resetting the Sr-Ca isotope composition, analogous to models proposed for other early dolomitization systems (Ahm et al., 2019). Direct hydrogeological evidence for this mechanism in the Gerze Basin remains limited, but the scenario reconciles Mg addition with the sediment-buffered isotopic signature of Dolomite EM1.

5.2. *Reconstructing lake-water composition and diagenetic fluid evolution*

The paragenetic sequence and end-member framework provide the basis for reconstructing both the primary lake-water composition and the subsequent evolution of diagenetic fluids. To quantify these components, the early-diagenetic model of Ahm et al. (2018) was adapted within a multi-proxy framework that combines $\delta^{18}O_{carb}$, $\delta^{44/40}Ca$, $^{87}Sr/^{86}Sr$ and Sr/Ca.

The high-Sr/Ca Calcite EM was used as the primary precursor phase. While the $\delta^{44/40}Ca$ values of potential diagenetic fluids in lacustrine successions, including lake water, groundwater, and river water, span a broad range from 0.00 ‰ to 4.00 ‰ (Schmitt, 2016), it is unlikely that the $\delta^{44/40}Ca$ value of a restricted lake is higher than 2.00 ‰, since such high values are generally associated with kinetic fractionation in alkaline lakes such as Mono Lake (Nielsen and DePaolo, 2013). Given regional hydrological $\delta^{44/40}Ca$ values of >~1.00 ‰ on the Tibetan Plateau (Tipper et al., 2006, 2008; Chen et al., 2022), the diagenetic fluid $\delta^{44/40}Ca$ value was assumed to be 1.5 ‰ in the numerical model. Sensitivity tests show that varying fluid $\delta^{44/40}Ca$ between 1.00 ‰ and 2.00 ‰ changes modeled diagenetic mineral $\delta^{44/40}Ca$ by only ~ 0.40 ‰, which cannot encompass the measured data (Fig. S11).

Model results require two distinct fluid conditions to reproduce the three end-members. Interaction between the high-Sr/Ca Calcite EM precursor and an ^{18}O-depleted diagenetic fluid ($\delta^{18}O_{VSMOW}$ = – 10.4 ‰) produces progressive Sr loss from Calcite EM at diagenetic degrees of 30–50 % and generates Dolomite EM2 at diagenetic degrees of 70–100 % (Fig. 8A). In contrast, Dolomite EM1 is reproduced by interaction between Calcite EM and a less ^{18}O-depleted fluid ($\delta^{18}O_{VSMOW}$ = – 4.1 ‰; Fig. 8B). These modeled fluid compositions are consistent with the interpretation that Dolomite EM1 formed from lake-derived porewaters under sediment-buffered conditions, whereas Dolomite EM2 formed during later interaction with a more evolved, ^{18}O-depleted diagenetic fluid.

Alternative precursor scenarios fail to reproduce the full Dolomite EM2 field. If Dolomite EM1 is treated as the primary mineral, most Dolomite EM2 compositions cannot be generated using fluids with $\delta^{18}O_{VSMOW}$ values of either – 10.4 ‰ or – 30.9 ‰ (Fig. 8C). Dolomite EM2 can be reproduced only if the primary dolomite has $\delta^{44/40}Ca$ values near 0.00 ‰, which is inconsistent with the measured calcium-isotope composition of Dolomite EM1. A mixed precursor composed of equal proportions of Calcite EM and Dolomite EM1 also fails to reproduce the full isotopic range of Dolomite EM2, even though this scenario represents the most favorable case for generating intermediate compositions between the two end-members (Fig. S12).

We further tested whether the Sr-isotope composition inferred from Dolomite EM2 can account for the observed $^{87}Sr/^{86}Sr$-$\delta^{44/40}Ca$ trend. The model uses the high-Sr/Ca Calcite EM precursor and assigns the

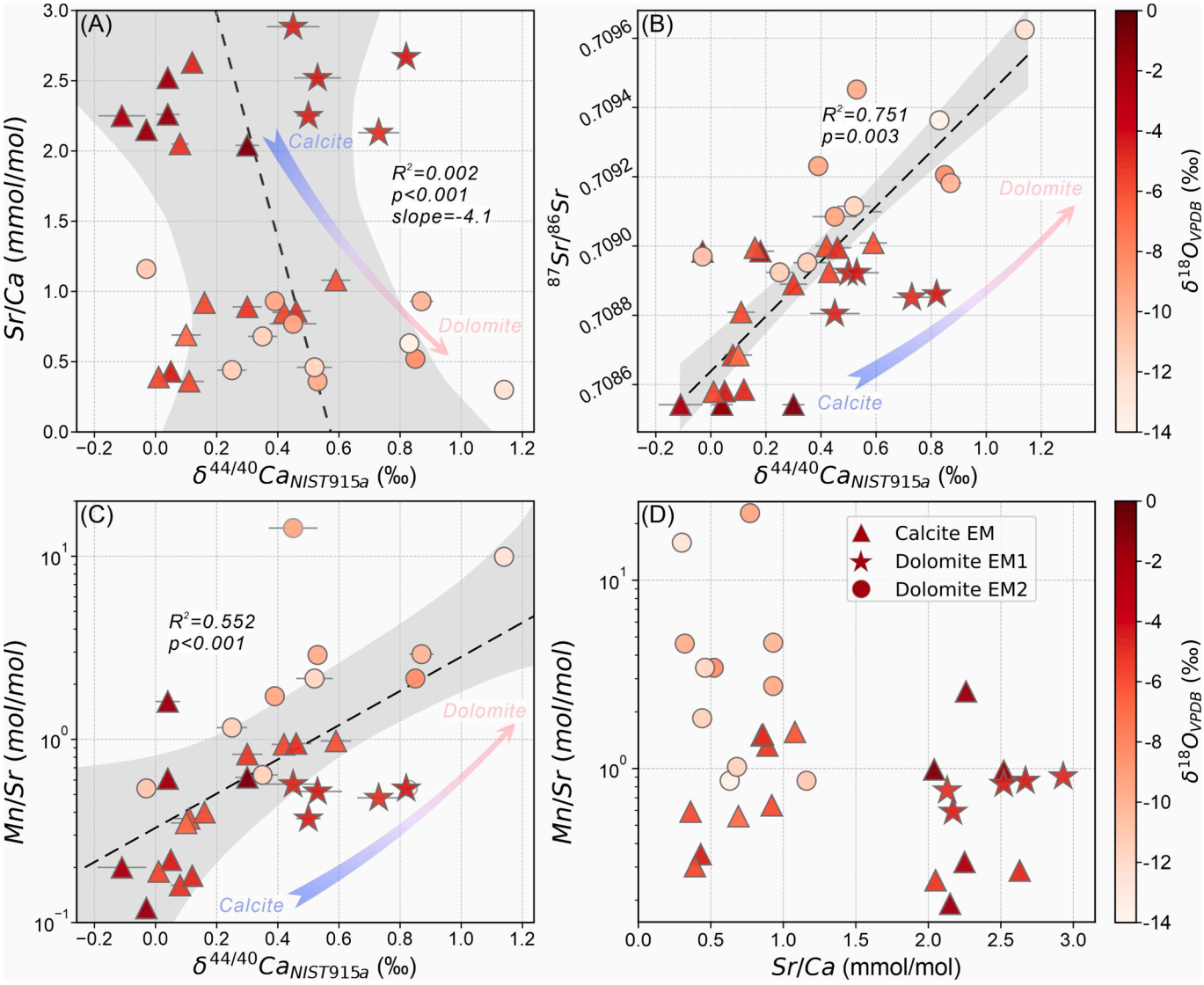


Fig. 7. Cross-plots of (A) $\delta^{44/40}$Ca vs. Sr/Ca, (B) $\delta^{44/40}$Ca vs. $^{87}Sr/^{86}Sr$, (C) $\delta^{44/40}$Ca vs. Mn/Sr, and (D) Mn vs. Sr.

shallow-burial diagenetic fluid $^{87}Sr/^{86}Sr$ value of 0.7096, based on the most radiogenic Dolomite EM2 compositions discussed in Section 5.1.2. Under these conditions, the model reproduces the shift from high-Sr/Ca Calcite EM toward low-Sr/Ca Calcite EM and Dolomite EM2 at diagenetic extents of 60–100 % (Fig. 8D). This agreement supports the interpretation that Dolomite EM2 formed through interaction with a more radiogenic shallow-burial fluid.

The modeled fluid compositions can then be combined with the paragenetic sequence to reconstruct lake-water $\delta^{18}O$. Because the high-Sr/Ca subset of Calcite EM represents the least altered carbonate phase, the $\delta^{18}O_{VSMOW}$ value of lake water was estimated from high Sr/Ca Calcite EM using calcite-water fractionation at near-surface temperature (~20 °C; Kim and O'Neil, 1997; see Supplementary Section S3). This calculation yields lake-water $\delta^{18}O_{VSMOW}$ values of − 4.28 ‰ to − 0.15 ‰. Ca and Sr isotopes provide complementary constraints on lake-water composition and subsequent fluid modification. Using a calcite-fluid calcium-isotope fractionation of $\Delta^{44/40}Ca_{calcite\text{-}fluid} \approx -0.90$ ‰ (Marriott et al., 2004; Gussone et al., 2005; Gussone and Dietzel, 2016), the corresponding lake-water $\delta^{44/40}$Ca value is estimated at ~ 1.0 ‰. Using a dolomite-fluid fractionation of $\Delta^{44/40}Ca_{dolomite\text{-}fluid} \approx -0.67$ ‰ (Wang et al., 2012; Blättler et al., 2015), the Stage 2 diagenetic fluid $\delta^{44/40}$Ca value is estimated at ~ 1.5 ‰. The $^{87}Sr/^{86}Sr$ ratio of Calcite EM (~0.7086) approximates the Sr isotopic composition of lake water. The higher $^{87}Sr/^{86}Sr$ ratio of Dolomite EM2 (~0.7096) therefore indicates addition of radiogenic Sr from an external source, plausibly linked to Qiangtang potassic volcanic rocks (0.710–0.712; Ding et al., 2003) or regional river waters on the Tibetan Plateau (~0.71694; Wu et al., 2009).

Together, the O-Ca-Sr isotope constraints define an initial lake-water end-member with $\delta^{18}O_{VSMOW}$ values of − 4.28 ‰ to − 0.15 ‰ and $\delta^{44/40}$Ca of ~ 1.0 ‰. Dolomite EM1 largely preserves this lake-derived porewater signal, whereas Dolomite EM2 records later modification by an evolved diagenetic fluid with lower $\delta^{18}O_{VSMOW}$ values, higher $\delta^{44/40}$Ca, and more radiogenic $^{87}Sr/^{86}Sr$. This distinction links the paragenetic sequence to a shift from Stage 1 lake-derived, sediment-buffered diagenesis to Stage 2 fluid-buffered alteration by externally influenced shallow-burial fluids. These reconstructed lake-water compositions indicate that the Stage 1 Gerze Basin was a semi-closed lake system with substantial evaporative enrichment. The calculated $\delta^{18}O_{VSMOW}$ values are more positive than local meteoric water but remain well below the extreme values typical of a semi-closed and evaporitic basin, and the ~ 4 ‰ range is consistent with a semi-closed lake hydrologic regime (Talbot, 1990; Leng and Marshall, 2004; Horton et al., 2016; Wu et al., 2017). This interpretation is consistent with the middle Eocene open-lake phase documented in the Gerze Basin (Jiang et al., 2016). The later shift toward lower $\delta^{18}O_{VSMOW}$, higher $\delta^{44/40}$Ca, and more

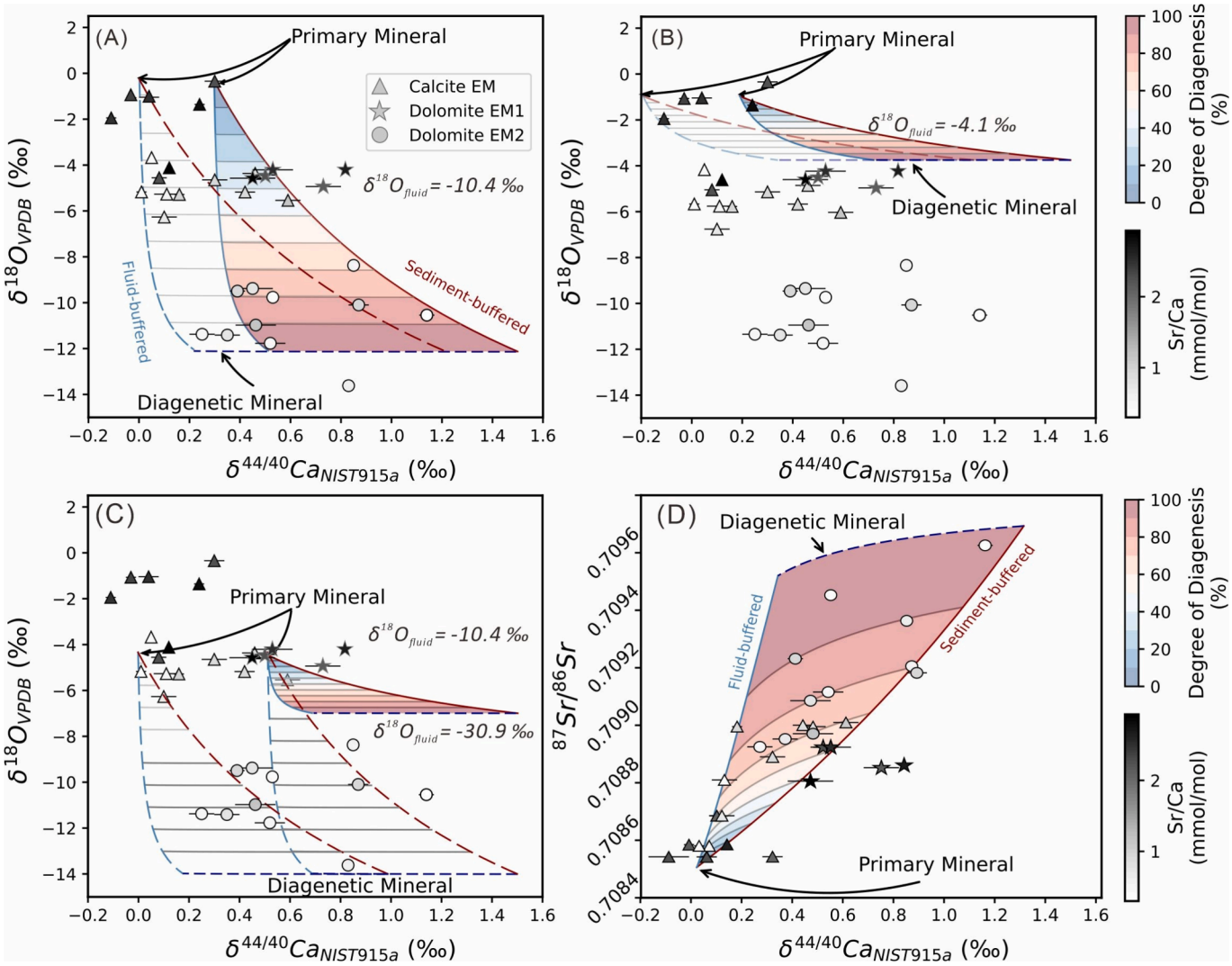


Fig. 8. Early lacustrine diagenesis model based on Ahm et al. (2018). The $\delta^{18}O_{fluid}$ is relative to VSMOW. (A) Primary mineral is set from high Sr/Ca Calcite EM ($\delta^{44/40}$Ca values of 0.0 ‰ or 0.4 ‰; $\delta^{18}O_{carb}$ value of − 1.0 ‰; $\delta^{18}O_{fluid}$ denotes diagenetic fluid oxygen isotope, following Coplen et al. (1983), which is − 10.4 ‰). (B) Primary mineral is set from high Sr/Ca Calcite EM ($\delta^{44/40}$Ca values of − 0.2 ‰ or 0.2 ‰; $\delta^{18}O_{carb}$ value of − 1.0 ‰; $\delta^{18}O_{fluid}$ value of − 4.1 ‰). (C) Primary mineral from Dolomite EM1 ($\delta^{44/40}$Ca values of 0.6 ‰; $\delta^{18}O_{carb}$ value of − 1.0 ‰; $\delta^{18}O_{fluid}$ value of − 10.4 ‰ or − 30.9 ‰) cannot form most observed Dolomite EM2 samples. Only primary dolomite with $\delta^{44/40}$Ca value of 0.0 ‰ can explain some observed samples. (D) $^{87}Sr/^{86}Sr$-$\delta^{44/40}$Ca modeling shows Calcite EM precursor ($^{87}Sr/^{86}Sr$ value of 0.7085; $\delta^{44/40}$Ca value of 0.0 ‰; $^{87}Sr/^{86}Sr_{fluid}$ value of 0.7096) could form two dolomitic end-members.

radiogenic $^{87}Sr/^{86}Sr$ in Dolomite EM2 further suggests continued access to external water sources during Stage 2, with enhanced meteoric or glacial recharge superimposed on an evaporative background (Pu et al., 2017).

5.3. Implications for paleo-elevation reconstruction

To evaluate biases introduced by different diagenetic carbonate end-members, we calculated paleo-elevation following Li et al. (2022), which improves on Ingalls et al. (2020) by incorporating vapor contributions from both northern and southern Tibet ($\delta^{18}O_{north}$ and $\delta^{18}O_{south}$). First, the $\delta^{18}O$ value of surface water was calculated from the $\delta^{18}O_{carb}$ values with a surface temperature assumption of 25 ± 10 °C (Li et al., 2022). Then the $\delta^{18}O_{south}$ was calculated by assuming that 36 % of the vapor originated from the south, with the remaining 64 % derived from $\delta^{18}O_{north}$ (Li and Garzione, 2017). Finally, paleo-elevation was determined using the Rayleigh distillation process (Li et al., 2022). The results show that Calcite EM yields paleo-elevations of around 1 km (Fig. 9). Because evaporative effects on meteoric-water $\delta^{18}O$ were not evaluated due to limited data, the calculated paleo-elevation represents a minimum estimate.

We also applied Monte Carlo methods to quantify the impact of end-member mixing. Assuming a mineral precipitation temperature of 25 ± 10 °C, commonly used in bulk-micrite paleo-elevation calculations (Wei et al., 2016), the reconstructed paleo-elevation from the bulk micrites is 4.1 ± 1.1 km (Fig. 9). Even when only sediment-buffered Dolomite EM1 and the least-altered calcite precursor Calcite EM are mixed (i.e., representing near-surface samples, NSS in Fig. 9), the reconstructed paleo-elevation is 2 km (Fig. 9), which serves as an approximate baseline for assessing the magnitude of diagenetic bias rather than a strict depositional constraint. The dominant mixing trend is between Calcite EM and the burial-diagenetic Dolomite EM2, which indicates that each 10 % increase in Dolomite EM2 contribution would increase apparent paleo-elevation by 0.5 to 0.8 km. This mixing effect highlights that even minor contributions from diagenetic end-members can cause over-estimation in the reconstruction of paleo-elevation.

These results challenge the assumption that micrite is a geochemically homogeneous primary phase suitable for direct paleo-elevation reconstruction. They further indicate that any studies of mixed calcite-dolomite micrites should combine mineral-specific separation with

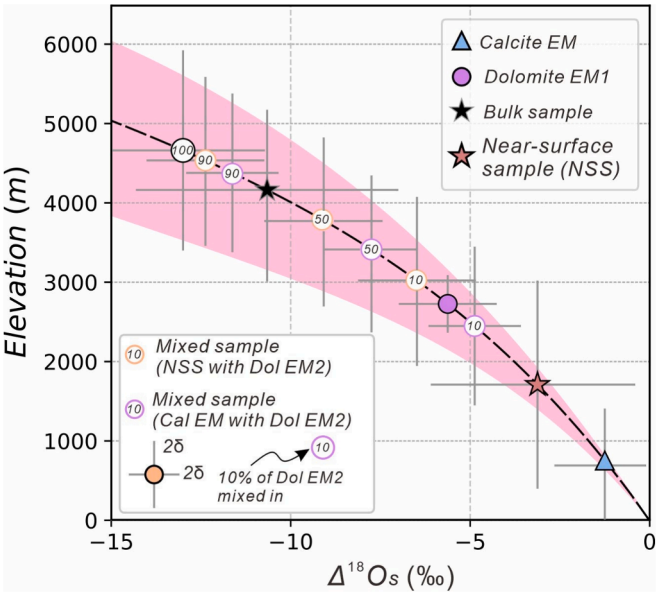


Fig. 9. Paleo-elevation reconstruction based on $\Delta^{18}O_s$, defined as the $\delta^{18}O$ difference between precipitation at the study site and a low-elevation reference site (New Delhi: −5.9 ‰; Li et al., 2022). The $\delta^{18}O$ of the near-surface sample (NSS, corresponding to Calcite EM with Dolomite EM1, shaded area) was estimated by Monte Carlo mixing of Calcite EM and Dolomite EM1 end-members. Numbers in circles along the mixing trajectories denote the proportion of Dolomite EM2 in the mixture (e.g., 10 indicates 10 % Dolomite EM2 mixed with 90 % Calcite EM or NSS). Error bars represent analytical and propagated uncertainties.

multi-proxy tracers (e.g., C-O-Sr-Ca isotopes and trace elements) to constrain both primary lake-water composition and diagenetic modification. This approach is applicable to orogenic lacustrine micritic carbonates in other basins, including the Lunpola and Nima basins in central Tibet (e.g., Rowley and Currie, 2006; Li et al., 2019; Fang et al., 2020; Li et al., 2022; Quade et al., 2020) and analogous basins in the Andes and Alpine-Himalayan belt (Jimenez-Rodriguez et al., 2022), provided that a primary calcite end-member is partly preserved and local diagenetic fluid compositions can be independently bracketed.

6. Conclusion

Multi-proxy isotopic and geochemical data show that the Eocene lacustrine micrites of the Gerze Basin consist of three carbonate end-members: a calcite end-member comprising near-primary and recrystallized calcite (Calcite EM), sediment-buffered dolomite (Dolomite EM1), and fluid-buffered dolomite (Dolomite EM2). Paragenetic and geochemical constraints indicate that Dolomite EM1 formed under near-surface, sediment-buffered conditions from lake-derived porewater with $\delta^{18}O_{VSMOW}$ values of about − 4.1 ‰, whereas Dolomite EM2 formed during later shallow-burial, fluid-buffered dolomitization from more ^{18}O-depleted fluids with $\delta^{18}O_{VSMOW}$ values of about − 10.4 ‰. Calcite EM preserves primary to near-surface lake-water signals and yield minimum paleo-elevations of approximately 1 km, whereas mixtures of Calcite EM and Dolomite EM1 representing near-surface samples (NSS) yield an apparent paleo-elevation of approximately 2 km and Dolomite EM2 systematically raises apparent paleo-elevation by 0.5–0.8 km for every 10 % increase in its contribution. Bulk micrite therefore inflates paleo-elevation estimates to 4.1 ± 1.1 km.

These results show that Gerze Basin micrite does not preserve a single homogeneous lake-water $\delta^{18}O$ signal. Previously reported bulk $\delta^{18}O_{carb}$ values likely reflect variable mixing between primary calcite, sediment-buffered dolomite, and fluid-buffered dolomite, rather than a direct surface-water record. Micrite from orogenic lacustrine basins should be evaluated through mineral-specific separation and multi-proxy diagenetic screening before being used for paleo-elevation reconstruction.

Data availability

Data are available through Mendeley Data at https://doi.org/10.17632/xkr7nvnv86.1. All code snippets in this study are accessible via https://github.com/KeranLi/diagenetic-fluid-overprinting.

CRediT authorship contribution statement

Keran Li: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anlin Ma:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Xiumian Hu:** Writing – review & editing, Supervision. **Chaojin Lu:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Yoseph Datu Adiatma:** Writing – review & editing, Validation, Software, Investigation, Data curation. **Lin Li:** Writing – review & editing, Formal analysis. **Ardiansyah Koeshidayatullah:** Writing – review & editing, Formal analysis. **He Zhang:** Formal analysis. **Di Yang:** Formal analysis. **Zhiwu Li:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study was supported by the Deep Earth Probe and Mineral Resources Exploration–National Science and Technology Major Project (grant 2025ZD1005402), the National Natural Science Foundation of China (Grant 42472161), State Key Laboratory of Critical Earth Material Cycling and Mineral Deposits (2026-Z05), the Fundamental Research Funds for the Central Universities, and State Key Laboratory of Oil and Gas Reservoir Geology and Exploitation (CDUT-PLC-2024013). CJL acknowledges support from National Natural Science Foundation of China (ZX20260130).

Appendix A. Supplementary material

The supplementary material comprises three sections, two tables and twelve figures: Phosphoric-acid fractionation calibration for carbonate $\delta^{18}O$ (Section S1); Reference material results for elemental and Sr-Ca analysis (Section S2); Fluid temperature analysis (Section S3); Parameters for the water–rock reaction burial model in this study (Table S1); Early lacustrine diagenesis modeling parameters (Table S2); Reference-material results for elemental concentrations, Sr isotopes, and Ca isotopes (Fig. S1); Results of the 5 % acetic acid leach experiment for selected samples from the study area (Fig. S2); Results of the 10 % acetic acid leach experiment for selected samples from the study area (Fig. S3); Results of the 15 % acetic acid leach experiment for selected samples from the study area (Fig. S4); Workflow of the sequential leaching (Fig. S5); Original EDS analysis results for sample DC14 (Fig. S6); Original EDS analysis results for sample DC26 (Fig. S7); Results of the two-end-member mixing model (Fig. S8); BayesMCMC modeling for four-end-member mixing (Fig. S9); Sensitivity analysis of BayesMCMC modeling for three-end-member mixing (Fig. S10); Simulation of different diagenetic fluid $\delta^{44/40}Ca$ (Fig. S11); An equal-proportion mixture of Calcite EM and Dolomite EM1 can explain some observed samples when the diagenetic fluid has $\delta^{18}O_{VSMOW} = -10.4$ ‰ or − 30.9 ‰ (Fig. S12). Supplementary material to this article can be found online

at https://doi.org/10.1016/j.gca.2026.09.026.

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